

$x_c(t) = A_c \cos[2\pi f_c t + \phi(t)]$	$f_i(t) = \frac{1}{2\pi} \frac{d\theta_i}{dt} = f_c + \frac{1}{2\pi} \frac{d\phi}{dt}$
$\phi(t) = k_p m(t)$	$d\phi(t)/dt = k_p dm(t)/dt$
$\frac{d\phi}{dt} = k_f m(t)$	$\phi(t) = k_f \int_{t_0}^t m(\alpha) d\alpha + \phi_0$
$x_{c_{PM}}(t) = A_c \cos[2\pi f_c t + k_p m(t)]$	$x_{c_{FM}}(t) = A_c \cos \left[2\pi f_c t + 2\pi f_d \int^t m(\alpha) d\alpha \right]$
$x_c(t) = \text{Re}(A_c e^{j\phi(t)} e^{j2\pi f_c t})$	$x_c(t) \cong A_c \cos(2\pi f_c t) - A_c \phi(t) \sin(2\pi f_c t)$
$f_c = nf_0 + f_{LO} \quad \text{or} \quad f_c = nf_0 - f_{LO}$	$x_c(t) = A_c \cos[2\pi f_c t + n\phi(t)]$
$M_1 \cdot M_2 = \frac{\Delta f_4}{\Delta f_1}$	$f_4 = M_1 M_2 f_{c1} \pm M_2 f_{LO}$
$y_D(t) = \frac{1}{2\pi} K_D \frac{d\phi}{dt}$	$y_D(t) = K_D f_d m(t)$
$y(t) = A_c \left(2\pi f_c + \frac{d\phi}{dt} \right)$	$f_c > -\frac{1}{2\pi} \frac{d\phi}{dt}$
$y_D(t) = A_c \frac{d\phi}{dt} = 2\pi A_c f_d m(t)$	$K_D = 2\pi A_c$
$x_c(t) = \text{Re}[\tilde{x}_c(t) e^{j2\pi f_c t}]$	$\tilde{x}_c(t) = A_c e^{j\beta \sin(2\pi f_m t)}$
$e^{j\beta \sin(2\pi f_m t)} = J_n(\beta) e^{j2\pi n f_m t}$	$x_c(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos[2\pi(f_c + n f_m)t]$
$J_{n+1}(\beta) = \frac{2n}{\beta} J_n(\beta) + J_{n-1}(\beta)$	$\beta_{PM} = k_p A$
$\beta_{FM} = \frac{k_f A}{2\pi f_m} = \frac{2\pi f_d A}{2\pi f_m} = \frac{f_d A}{f_m}$	$P_r = J_0^2(\beta) + 2 \sum_{n=1}^k J_n^2(\beta)$
$B = 2 k f_m \text{ Hz}$ $P_r \geq 0.98, k = \beta + 1.$	$D = \frac{\text{peak freq. deviation}}{W} = \frac{f_d}{W} \text{ at } \max(m(t))$
$B \cong 2(D + 1) W \text{ Hz}$ carson rule	$B = 2f_d \text{ at } \max(m(t))$
$\bar{g}(t) = g(t) \delta_{T_s}(t) = \sum_n g(nT_s) \delta(t - nT_s)$	$\bar{g}(t) = g(t) \delta_{T_s}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} g(t) e^{jn2\pi f_s t}$

$\bar{G}(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} G(f - nf_s)$	$\Delta v = \frac{2m_p}{L} \quad N_q = \overbrace{q^2(t)} = \frac{m_p^2}{3L^2}$
$\frac{S_o}{N_o} = 3L^2 \frac{m^2(t)}{m_p^2}$	$n = \lceil \log_2(L) \rceil \text{ bits} \quad B_T = nB \text{ Hz}$
$\frac{S_o}{N_o} = c(2)^{2n}$ $\left(\frac{S_o}{N_o}\right)_{\text{dB}} = 10 \log_{10} \left(\frac{S_o}{N_o}\right)$ $= (\alpha + 6n) \text{ dB}$	$c = \begin{cases} \overbrace{3 \frac{m^2(t)}{m_p^2}} \\ \frac{3}{[\ln(1 + \mu)]^2} \end{cases}$
$N_q = (\Delta v)^2 / 12$	$\frac{S_o}{N_o} \simeq \frac{3L^2}{[\ln(1 + \mu)]^2} \quad \mu^2 \gg \frac{m_p^2}{m^2(t)}$

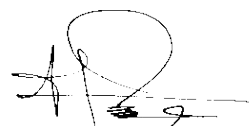
TABLE 3.1
Short Table of Fourier Transforms

$g(t)$	$G(f)$
1 $e^{-at}u(t)$	$\frac{1}{a + j2\pi f}$
2 $e^{at}u(-t)$	$\frac{1}{a - j2\pi f}$
3 $e^{-a t }$	$\frac{2a}{a^2 + (2\pi f)^2}$
4 $te^{-at}u(t)$	$\frac{1}{(a + j2\pi f)^2}$
5 $t^n e^{-at}u(t)$	$\frac{n!}{(a + j2\pi f)^{n+1}}$
6 $\delta(t)$	1
7 1	$\delta(f)$
8 $e^{j2\pi f_0 t}$	$\delta(f - f_0)$
9 $\cos 2\pi f_0 t$	$0.5 [\delta(f + f_0) + \delta(f - f_0)]$
10 $\sin 2\pi f_0 t$	$j0.5 [\delta(f + f_0) - \delta(f - f_0)]$
11 $u(t)$	$\frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$
12 $\text{sgn } t$	$\frac{2}{j2\pi f}$
13 $\cos 2\pi f_0 t u(t)$	$\frac{1}{4} [\delta(f - f_0) + \delta(f + f_0)] + \frac{j2\pi f}{(2\pi f_0)^2 - (2\pi f)^2}$
14 $\sin 2\pi f_0 t u(t)$	$\frac{1}{4j} [\delta(f - f_0) - \delta(f + f_0)] + \frac{2\pi f_0}{(2\pi f_0)^2 - (2\pi f)^2}$
15 $e^{-at} \sin 2\pi f_0 t u(t)$	$\frac{2\pi f_0}{(a + j2\pi f)^2 + 4\pi^2 f_0^2}$
16 $e^{-at} \cos 2\pi f_0 t u(t)$	$\frac{a + j2\pi f}{(a + j2\pi f)^2 + 4\pi^2 f_0^2}$

Properties of the Fourier Transform

Operation	$g(t)$	$G(f)$
Superposition	$g_1(t) + g_2(t)$	$G_1(f) + G_2(f)$
Scalar multiplication	$kg(t)$	$kG(f)$
Duality	$G(t)$	$g(-f)$
Time scaling	$g(at)$	$\frac{1}{ a } G\left(\frac{f}{a}\right)$
Time shifting	$g(t - t_0)$	$G(f)e^{-j2\pi f t_0}$
Frequency shifting	$g(t)e^{j2\pi f_0 t}$	$G(f - f_0)$
Time convolution	$g_1(t) * g_2(t)$	$G_1(f)G_2(f)$
Frequency convolution	$g_1(t)g_2(t)$	$G_1(f) * G_2(f)$
Time differentiation	$\frac{d^n g(t)}{dt^n}$	$(j2\pi f)^n G(f)$
Time integration	$\int_{-\infty}^t g(x) dx$	$\frac{G(f)}{j2\pi f} + \frac{1}{2}G(0)\delta(f)$

Associate Prof. Dr. Ali Kadhim Al-Janabi

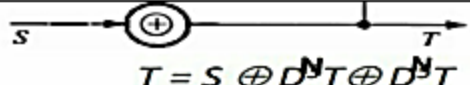


17	$\Pi\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc}(\pi f \tau)$
18	$2B \operatorname{sinc}(2\pi B t)$	$\Pi\left(\frac{f}{2B}\right)$
19	$\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2} \operatorname{sinc}^2\left(\frac{\pi f \tau}{2}\right)$
20	$B \operatorname{sinc}^2(\pi B t)$	$\Delta\left(\frac{f}{2B}\right)$
21	$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$f_0 \sum_{n=-\infty}^{\infty} \delta(f - nf_0)$
22	$e^{-t^2/2\sigma^2}$	$\sigma \sqrt{2\pi} e^{-2(\sigma\pi f)^2}$

TABLE 3.3

$E_g = \int_{-\infty}^{\infty} g^2(t) dt$	$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt = \lim_{T \rightarrow \infty} \frac{E_{gT}}{T}$
$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g(t+\tau) dt$	$\mathcal{R}_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g(t+\tau) dt = \lim_{T \rightarrow \infty} \frac{\psi_{gT}(\tau)}{T}$
$\Psi_g(f) = G(f) ^2$	$S_g(f) = \lim_{T \rightarrow \infty} \frac{ G_T(f) ^2}{T} = \lim_{T \rightarrow \infty} \frac{\Psi_{gT}(f)}{T}$
$\psi_g(\tau) \Leftrightarrow \Psi_g(f)$	$\mathcal{R}_g(\tau) \Leftrightarrow S_g(f)$
$E_g = \int_{-\infty}^{\infty} \Psi_g(f) df$	$P_g = \int_{-\infty}^{\infty} S_g(f) df$

$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$	$g(t) = \int_{-\infty}^{\infty} G(f) e^{j\omega t} df$
$\operatorname{sinc}(x) = \frac{\sin x}{x}$	$\operatorname{sgn} t = \lim_{a \rightarrow 0} [e^{-at} u(t) - e^{at} u(-t)]$
$E_{\psi} = \frac{1}{2} E_g \quad f_0 \geq B$	$P_{\psi} = \frac{1}{2} P_g \quad f_0 \geq B$
$m(t) \cos 2\pi f_c t \Leftrightarrow \frac{1}{2} [M(f + f_c) + M(f - f_c)]$ $\eta = \frac{\text{useful power}}{\text{total power}} = \frac{P_s}{P_c + P_s} = \frac{m^2(t)}{A^2 + m^2(t)} 100\%$ $f_c \gg B$	$\varphi_{AM}(t) = A \cos \omega_c t + m(t) \cos \omega_c t$ $= [A + m(t)] \cos \omega_c t$ $[A + m(t)] \geq 0$
$\mu = \frac{m_{\max} - m_{\min}}{2A + m_{\max} + m_{\min}}$ $0 \leq \mu \leq 1$ $\mu = \frac{m_p}{A}$ $A \geq m_p$	$P_c = \frac{A^2}{2}$ and $P_s = \frac{1}{2} m^2(t)$ $\eta = \frac{\mu^2}{2 + \mu^2} 100\%$
$\varphi_{SSB}(t) = \cos \omega_m t \cos \omega_c t$ $= \cos(\omega_c \pm \omega_m)t$ $\varphi_{USB}(t) = \cos(\omega_c + \omega_m)t$ and $\varphi_{LSB}(t) = \cos(\omega_c - \omega_m)t$	$m_h(t) = m(t) * \frac{1}{\pi t}$ $M_h(f) = -j \operatorname{sgn}(f) M(f)$ $H(f) = -j \operatorname{sgn}(f)$ $= \begin{cases} -j = 1 \cdot e^{-j\pi/2} & f > 0 \\ j = 1 \cdot e^{j\pi/2} & f < 0 \end{cases}$
$y(t) = h(t) * x(t) \quad Y(f) = H(f) \cdot X(f)$ $H(f) = H(f) e^{j\theta_h(f)}$ $ Y(f) e^{j\theta_y(f)} = X(f) H(f) e^{j[\theta_x(f) + \theta_h(f)]}$	$\varphi_{USB}(t) = m(t) \cos \omega_c t - m_h(t) \sin \omega_c t$ $\varphi_{LSB}(t) = m(t) \cos \omega_c t + m_h(t) \sin \omega_c t$ $\Phi_{USB} = M_+(f - f_c) + M_-(f + f_c)$
$\varphi_{SSB+C} = A \cos \omega_c t + [m(t) \cos \omega_c t + m_h(t) \sin \omega_c t]$ $A \gg m(t) $	$\varphi_{QAM}(t) = m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t$
$x_2(t) = 2\varphi_{QAM}(t) \sin \omega_c t = 2[m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t] \sin \omega_c t$ $= m_2(t) - m_2(t) \cos 2\omega_c t + m_1(t) \sin 2\omega_c t$	$x_1(t) = 2\varphi_{QAM}(t) \cos \omega_c t = 2[m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t] \cos \omega_c t$ $= m_1(t) + m_1(t) \cos 2\omega_c t + m_2(t) \sin 2\omega_c t$

$E_g = \int_{-\infty}^{\infty} g^2(t) dt$	$y(t) = \sum a_k p(t - kT_b)$
$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$	$R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$
$\int_{-\infty}^{\infty} \phi(t) \delta(t - T) dt = \phi(T) \int_{-\infty}^{\infty} \delta(t - T) dt = \phi(T)$	$R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k^2 = \overline{a_k^2}$
$c = \frac{\int_{t_1}^{t_2} g(t)x(t) dt}{\int_{t_1}^{t_2} x^2(t) dt} = \frac{1}{E_x} \int_{t_1}^{t_2} g(t)x(t) dt$	$S_y(f) = P(f) ^2 S_x(f)$ $= \frac{ P(f) ^2}{T_b} \left[R_0 + 2 \sum_{n=1}^{\infty} R_n \cos n2\pi f T_b \right]$
$C_n = \frac{1}{\sqrt{E_g E_z}} \int_{-\infty}^{\infty} g(t) z(t) dt$	$S_x(f) = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} R_n e^{-jn2\pi f T_b}$
$\rho = \frac{1}{\sqrt{E_g E_z}} \int_{-\infty}^{\infty} z(t) g^*(t) dt = 0$	Thus $B_s = 2R_b = 4 B_N$ Hz For a full-width pulse (NRZ pulse) $B_s = R_b = 2B_N$ Hz
$\psi_{gz}(\tau) \equiv \int_{-\infty}^{\infty} z(t) g^*(t - \tau) dt = \int_{-\infty}^{\infty} z(t + \tau) g^*(t) dt$	 $T = S \oplus D^N T \oplus D^N T$
$\psi_g(\tau) \equiv \int_{-\infty}^{\infty} g(t) g(t + \tau) dt$	$S = T \oplus D^N T \oplus D^N T$ $\varphi_{PSK}(t) = m(t) \cos \omega_c t \quad m(t) = \sum a_k p(t - kT_b)$
$D_n = \frac{1}{T_0} \int_{T_0} g(t) e^{-jn2\pi f_0 t} dt$	$\varphi_{FSK}(t) = \sum a_k p(t - kT_b) \cos \omega_{c1} t + \sum (1 - a_k) p(t - kT_b) \cos \omega_{c2} t$
$P_g = D_0^2 + 2 \sum_{n=1}^{\infty} D_n ^2$	$I_M = \log_2(M)$ bits
$\phi(t) \delta(t - T) = \phi(T) \delta(t - T)$	$\varphi_{FSK}(t) = A \cos \theta_k \cos \omega_c t - A \sin \theta_k \sin \omega_c t$ $= a_k \cos \omega_c t + b_k \sin \omega_c t \quad kT_b \leq t < kT_b + T_b$

Half Angle Formulas

$$\sin^2 \theta = \frac{1}{2} (1 - \cos(2\theta))$$

$$\cos^2 \theta = \frac{1}{2} (1 + \cos(2\theta))$$

$$\tan^2 \theta = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}$$

Sum and Difference Formulas

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Product to Sum Formulas

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$$

$$\sin A = \frac{e^{jA} - e^{-jA}}{2j}$$

$$\cos A = \frac{e^{jA} + e^{-jA}}{2}$$

Handwritten signature

TABLE 3.1 TABLE OF BESSEL FUNCTION VALUES

n	$\beta = 0.1$	$\beta = 0.2$	$\beta = 0.5$	$\beta = 1$	$\beta = 2$	$\beta = 5$	$\beta = 8$	$\beta = 10$
0	0.997	0.990	0.938	0.765	0.224	-0.178	0.172	-0.246
1	0.050	0.100	0.242	<u>0.440</u>	<u>0.577</u>	-0.328	0.235	0.043
2	0.001	0.005	0.031	<u>0.115</u>	0.353	0.047	-0.113	0.255
3				<u>0.020</u>	<u>0.129</u>	0.365	-0.291	0.058
4				0.002	<u>0.034</u>	<u>0.391</u>	-0.105	-0.220
5					0.007	0.261	0.186	-0.234
6					0.001	<u>0.131</u>	0.338	-0.014
7						<u>0.053</u>	<u>0.321</u>	0.217
8						0.018	<u>0.223</u>	<u>0.318</u>
9						0.006	<u>0.126</u>	<u>0.292</u>
10						0.001	<u>0.061</u>	0.207
11							0.026	<u>0.123</u>
12							0.010	0.063
13							0.003	0.029
14							0.001	0.012
15								0.004
16								0.001

(From Ziemer and Tranter; © 1990 Houghton Mifflin, reprinted by permission.)